EFU LIVESTOCK LICENCE v1.0

Experimental EFU-based Livestock & Sovereignty Framework

0. Purpose and Scope

Experimental EFU-based licensing and measurement framework for livestock farming (pigs, poultry, cattle, aquaculture).
NOT a legal breeding permit, nitrates directive, or organic certification replacement, but an open research protocol measuring livestock resource use in human-scale EFU metrics (feed, methane, water, soil loading).

I. Core Concepts

1. **EFU-Livestock (EFU-LIVESTOCK)**
   EFU framework application for livestock:
   - **EFU-FEED**: feed conversion (feed conversion ratio)
   - **EFU-CH4**: methane emissions (enteric fermentation)
   - **EFU-N**: nitrogen (manure, ammonia)
   - **EFU-W**: water (drinking, feed irrigation, cleaning)
   - **EFU-SOIL**: pasture degradation + manure loading
2. **Livestock Sovereignty Gap (SS_LIVESTOCK)**
   SS_LIVESTOCK = TEFU_LIVESTOCK_total – Regional Fertility Carrying Capacity (RFCC_LIVESTOCK)
   Measures if livestock stays within pasture/feedland regeneration capacity.
3. **MROI_LIVESTOCK**
   MROI_LIVESTOCK = THI_LIVESTOCK / TEFU_LIVESTOCK
   THI_LIVESTOCK = protein/calorie yield + local food security + animal welfare.

II. EFU Livestock Licence – Research Status

Documents EFU-audited livestock farms/operations.
Does NOT replace animal welfare, hygiene, or environmental authority permits.
Enables pig, poultry, cattle farm benchmarking.

III. Metrics and Thresholds

1. EFU Components (per kg meat or liter milk)

EFU-FEED
EFU-FEED = feed_kg / 300kg/person/year / meat_kg
Example: Chicken FCR=1.8 → 1.8/300/1 = 0.006 EFU/kg

EFU-CH4
EFU-CH4 = methane_kg / 0.2t/person/year / meat_kg
Example: Beef = 100kg CH4/year → high CH4 factor

EFU-N
EFU-N = nitrogen_emissions_kg / 50kg/person/year / meat_kg

TEFU_LIVESTOCK = 1.5×EFU-FEED + 2.0×EFU-CH4 + 1.5×EFU-N + 1.0×EFU-W + 1.2×EFU-SOIL

2. Regional Fertility Carrying Capacity (RFCC_LIVESTOCK)

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RFCC_LIVESTOCK = Sustainable_meat_kg / Population / 50kg/person/year baseline
Example: Ireland = 700k t / 5M / 50kg = 2.8 EFU_LIVESTOCK/person/year
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IV. Licence Categories (EFU-LIVESTOCK Labels)

Level	SS_LIVESTOCK	MROI_LIVESTOCK	Label	Interpretation
1	≤ 0	≥1e-2	**EFU-LIVESTOCK SOVEREIGN**	Within pasture capacity
2	0 < ≤15k EFU	≥5e-3	**EFU-LIVESTOCK BALANCED**	Monitorable
3	15k < ≤75k EFU	≥1e-3	**EFU-LIVESTOCK DEFICIT**	Feed dependency
4	>75k EFU	<1e-3	**EFU-LIVESTOCK CRITICAL**	Metabolic overload

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V. Regional RFCC_LIVESTOCK Calibrations

Country	Meat (k t/year)	Population (M)	RFCC_LIVESTOCK (EFU/person/year)
IRELAND	700	5	2.80
USA	40,000	340	0.24
BRAZIL	15,000	215	0.14
CHINA	80,000	1,410	0.11
EU	25,000	448	0.11

VI. Example: Irish Beef Finishing Farms

Input (annual sector):

- Meat: 700,000 tonnes
- Feed: 3.5M tonnes
- CH4: 2.5M tonnes
- N manure: 500,000 tonnes
- Workers: 50,000 staff
- Region: Ireland (RFCC_LIVESTOCK = 2.8 EFU/person/year)

Calculation:

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EFU-FEED = 3.5M/300/700k = 0.0167 EFU/kg
EFU-CH4 = 2.5M/0.2/700k = 0.0179 EFU/kg
EFU-N = 500k/50/700k = 0.0143 EFU/kg

TEFU_LIVESTOCK_total ≈ 42,000,000 EFU/year
Allocated = 50k × 2.8 = 140,000 EFU/year
SS_LIVESTOCK = 42M – 140k = +41,860,000 EFU/year (Level 4: CRITICAL)

Result: EFU-LIVESTOCK CRITICAL
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VII. Implementation Notes

1. **CH4 tax basis**: TEFU-CH4 metric for methane accounting
2. **Nitrates Directive**: EFU-N indicator for manure application regulation
3. **Data sources**: Tier 1 = feed bills, manure logs; Tier 2 = FAO statistics